Comparing Arabic Transformer Models for Question Answering Task

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***Abstract*—Natural Language Processing (NLP) is a growing field of artificial intelligence (AI) that focuses mainly on enabling computers to understand and analyze human natural language. Natural Language Processing (NLP) bridges the communication gap between computers and humans. This emergence leads to the development of several important applications that are used widely in today's life, including sentiment analysis that analyzes the emotions behind opinions, text summarization that generates a short version of a specific text, and question answering. Question answering (QA) task is mainly concerned with providing relevant answers based on the user's questions or extracting the answer from a given context. Despite the rapid growth of NLP in English, Arabic NLP is progressing slowly due to several constraints, such as the difficulty of the language and the availability of high-quality annotated datasets. This research contributes to the Arabic NLP by offering a comprehensive comparison of the existing pre-trained Arabic transformer-based models, namely MARBERTv2 and CAMeLBERT, on a several benchmark datasets for the extractive QA task.**

***Index Terms*—Extractive Question Answering, Arabic NLP, Transformer Models, Pre-trained Language Models, MARBERTv2, CAMeLBERT, BERT-based Models, TyDiQA-GoldP(Arabic), ARCD, AQAD, Arabic-SQuAD.**

I. Introduction

Natural Language Processing (NLP) is a branch of artificial intelligence that aims to make computers understand and produce human language. It has numerous applications, including machine translation, virtual assistants, information retrieval systems, chatbots, and automated text analysis, making NLP an integral part of today's world. NLP systems need machine-readable text, and large samples of text are easily obtained from websites such as Kaggle, HuggingFace Datasets, and other open sources. However, the rich morphology, many dialects, complex grammar, and diglossia (MSA and regional dialects) combine to make Arabic distinctly challenging. As a result, NLP applied to specialized areas, such as question answering (QA), becomes a daunting task.

The purpose of this project is to apply NLP techniques, including transfer learning, text preprocessing, and transformer-based modeling, to fine-tune and evaluate Arabic models for question answering (QA) over text. In this project, we wish to analyze and evaluate the performance of two Arabic pre-trained transformer models, MARBERTv2 and CAMeLBERT, on Arabic QA datasets obtained from HuggingFace, including TyDiQA-GoldP(Arabic), ARCD, AQAD, and Arabic-SQuAD. These models are of particular interest because each epitomizes one of the two broad, disparate approaches to modeling the Arabic language. MARBERTv2 specializes in the dialect and informal text because it is trained on 1B Arabic tweets and is thus suited to the dialects.CAMeLBERT enables the evaluation of the effect of language variation on downstream QA performance since it is trained on several models of Arabic, including Modern Standard Arabic (MSA), dialectal Arabic, and classical Arabic.

II. Project Goals and Objectives

This project aims to evaluate the performance of pre-trained Arabic transformer models on several benchmark datasets to gain a clear understanding of their generalization ability across different text domains and question types. Through this study, we aim to determine which model achieves superior performance for Arabic extractive question answering (QA) task, explore different types of benchmarking datasets, and contribute to Arabic NLP by offering a performance comparison between MARBERTv2 and CAMeLBERT.

III. Related Works (Literature Review)

Paper [1] gives a full overview of Arabic question answering systems. It looks at 26 different research papers and six main benchmark datasets. The writers focus on the main steps in these systems. Question analysis comes first. Then information retrieval. Answer extraction follows that. They point out that Arabic QA falls way behind other languages. This happens because of few language resources. Plus there is too much use of handmade methods. The review finds that most systems train on tiny datasets. Many of those datasets stay private and not open to everyone. Systems often use machine translated texts too. That brings in errors and hurts how well models work. The paper points out the missing use of deep learning in Arabic QA. Yet deep learning works great in English QA studies. In the end this review shows big problems. Datasets are not enough. Advanced neural setups see little use. There are no standard ways to check results. Answer pulling gets little attention. All this calls for more work on big Arabic QA collections. It also pushes for deep learning approaches. Limitations include no new experiments. The work just reviews other papers. Validated datasets in the area are hard to find. Techniques mentioned cover rule based methods. Hybrid machine learning models appear too. Basic natural language processing steps show up. That means tokenization and stemming. Named entity recognition fits in there. Vector space model information retrieval comes up as well.

Paper [2] really zeros in on the question analysis part of Arabic QA systems. It covers things like question classification, tokenization, entity extraction, and query expansion. The work looks at both old school machine learning approaches and the newer deep learning ones. In the end, it points out that deep learning does a better job with the tricky parts of Arabic. Still, traditional ML holds up well for basic classification stuff. The paper points to some big hurdles in the language. Those include its heavy morphology, unclear syntax at times, different dialects, and not enough NLP resources around. It pushes for standard ways to evaluate these systems. Those should use measures like precision, recall, F1 score, MRR, MAP, and C at 1. The writers stress how important it is to get shared datasets in place. They also want improved tools for preprocessing the language. Plus, they suggest mixing rule based methods with neural networks in pipelines. That way, question understanding gets stronger overall. The survey has some limits though. It sticks to just the question analysis area. The reviews of systems come across as more descriptive than hands on testing. The whole field still misses out on solid benchmarks to compare against. Some key techniques show up in the discussion. Those are models based on BERT, ELECTRA setups, ways that blend rules with semantics, named entity recognition, dependency parsing, and various embeddings.

Paper [3] introduces Hajj-FQA, a benchmark dataset specifically created to support question-answering over Islamic fatwas related to Hajj. The dataset includes carefully curated and annotated fatwa texts and questions collected from verified religious authorities. It supports multiple QA tasks, including machine reading comprehension, duplicate-question detection, and semantic answer similarity. The authors evaluate multiple language models—from classical Arabic models to advanced LLMs— highlighting the difficulty of handling religious language, long-context reasoning, and domain-specific terminology. Results demonstrate that while transformer models show promising performance, there is still substantial room for improvement in capturing nuanced religious rulings, context, and semantic equivalence in Arabic. Limitations: Domain-specific dataset; results may not generalize beyond religious QA; requires religious experts for annotation, slowing expansion. Techniques Mentioned: Transformer-based models (BERT variants, DeBERTa, AraBERT), semantic similarity scoring, MRC pipelines.

Paper [4] studies 40 Arabic QA papers published between 2013–2022. The review identifies research trends, classifies methods (rule-based, ML, deep learning), and surveys publicly available datasets. The study finds that most Arabic QA efforts target open-domain QA, yet focus heavily on document retrieval instead of answer extraction, which is essential for deep comprehension. The authors confirm severe dataset bottlenecks—many datasets are very small, unpublished, or translated from English—leading to weak model generalization. They also identify a lack of reading-comprehension-oriented models compared to English QA developments such as SpanBERT and SPLINTER. The paper recommends developing larger, high-quality native Arabic datasets, adopting transformer-based models, and prioritizing answer extraction evaluation. Limitations: Does not present new models or experiments;limited to published literature; dependent on existing public documentation.Techniques: Mentioned: Taxonomy of rule-based vs. deep learning techniques,Qur'anic datasets, transformer baselines, IR-centric pipelines.

Paper [5] presents a comprehensive survey of Arabic QA systems. It explains that the rapid growth of Arabic content on the internet has made the standard search engines inadequate for delivering answers with precision. Therefore, the researchers began developing QA systems that rely on Natural Language Processing and information retrieval methodologies. It describes the major types of Arabic QA systems: rule-based, retrieval-based, and hybrid approaches. The paper explains how these systems process user questions, retrieve relevant passages, and extract answers. According to the authors, Arabic QA lags far behind English QA because of the scarcity of annotated datasets, fewer linguistic tools, and more complicated morphology in Arabic. This survey reviews some of the existing Arabic QA systems according to methodology, question type, pipeline design, accuracy, and datasets used. It concludes that most of the Arabic QA systems heavily rely on surface text processing and lack deep semantic understanding-primarily due to language-specific challenges such as missing diacritics, morphological richness, and a scarcity of resources. The authors identify future needs in the creation of large datasets, improvements of NLP tools, development of modern machine-learning-based QA models, and construction of standardized evaluation metrics. The limitations of this study are the following: Limited quantity of Arabic QA datasets, weak NLP tool support for Arabic morphology and syntax, heavy reliance on handcrafted rules, and no unified benchmarks for fair comparison.

Paper [6] presents the first end-to-end Machine Reading at Scale system for question answering from the Holy Qur'an. The system follows a two-tier architecture: a retriever that selects relevant Qur'anic passages and a reader that extracts answers from the selected passages. It contributes significantly with QRCD, the first extractive Qur'anic Reading Comprehension Dataset with 1,337 question-passage-answer triplets. The system faces a significant linguistic challenge. While the questions are written in MSA, the Qur'an is written in CA. The author therefore develops CL-AraBERT, a specialized version of AraBERT that is further pre-trained on more than 1.05 billion words of Classical Arabic. The retriever uses BM25 with document expansion to enhance recall. The reader uses CL-AraBERT fine-tuned on the MSA and CA datasets. Experimental results show that the expanded retriever significantly improves the baseline, and CL-AraBERT improves the answer extraction accuracy significantly. The performance is notably stronger on single-answer questions than on multi-answer questions. It also proposes a new evaluation metric, Partial Average Precision (pAP), tailored for extractive multi-answer questions. The limitations of this paper are the following: Multi-answer questions remain challenging, the dataset size is relatively small, BERT-based extractive models do not naturally support multi-span answers, and limited ability to handle reasoning-heavy queries.

Paper [7] describes the QCRI submission to SemEval-2015 Task 3 that selects the best answers in community question-answering forums for Arabic and English. The Arabic part is based on Fatwa forums and the English part is based on data from Qatar Living. It uses supervised machine learning on a large suite of features, including lexical similarity using n-grams and cosine similarity; syntactic similarity based on partial tree kernels; semantic similarity using word embeddings; sentiment features; and contextual cues from surrounding comments. Currently, for the Arabic language, it classifies comments into Direct, Related, or Irrelevant, while comments in the English language are classified as Good, Potential, or Bad. The system performed best in the Arabic subtask and ranked among the top three systems in the English subtasks. The authors demonstrate that combining syntactic, semantic, and lexical features greatly helps to identify useful comments. The limitations of the paper are the following: Heavy dependence on feature engineering, quality depends strongly on the performance of linguistic tools, and label categories are broad and may oversimplify comment usefulness.

Paper [8] addresses the task of ranking similar questions in Arabic community forums; this is a key challenge in preventing duplicate posts and improving user search experience. The authors build an advanced Arabic NLP pipeline based on the Farasa toolkit and integrate it into a UIMA framework. The system uses Support Vector Machines that combine several sources of information: tree kernels applied to constituency parse trees, word embeddings for semantic similarity, and traditional similarity measures. In order to reduce the noise of the long user-generated questions, the authors introduce an LSTM-based attention mechanism that is able to automatically highlight the most relevant text segments before the application of structural kernels. Experiments demonstrate that Farasa's parsing and segmentation yield significant improvement in the accuracy of syntactic features. Moreover, the combination of tree kernels with deep learning-based attention improves speed and ranking performance. This model constantly outperforms IR-based baselines in retrieving semantically similar forum questions. The limitations of the paper are the following: Parsing errors may affect accuracy, especially with dialectal Arabic, kernel-based models are computationally costly, and user-generated content is highly variable, affecting generalization.

Paper [9] provided a detailed overview of the existing approaches for Arabic Question Answering (QA) systems. They distinguished three broad categories for Arabic QA mechanisms: closed-domain, open-domain, and reading comprehension systems. The authors focused on the complete process of system design, including the steps of preprocessing, question classification, information retrieval, and answer extraction systems. They pointed toward important issues related to dialectal variation, Arabic morphology, and a gap in published annotated datasets to construct QA systems. In addition, the survey called out issues caused by a lack of standardized quality datasets and evaluation metrics, which affect the ability to fairly review Arabic QA models. This work demonstrates a strong stepping-off point for identifying challenges within Arabic QA and future efforts for model design or dataset development.

Paper [10] presented ArabicaQA, one of the largest and most diverse Arabic QA datasets available today that offers nearly 89 K answerable and 3.7 K unanswerable questions collected using large-scale human judgment. The authors also proposed AraDPR, a dense passage retrieval model designed specifically for Arabic text. Extensive experimentation was conducted evaluating several large language model (LLM) architectures on machine reading comprehension (MRC) and open-domain QA tasks. Demonstrating improvement in both retrieval quality and overall answer accuracy with the proven benchmark of ArabicaQA dataset, the foundation of their work provides significant resources for future studies that aim to improve Arabic QA performance using retrieval and transformer-based models.

Paper [11] on WikiTableQA in Arabic research represents a new area of Arabic QA, which concerns structured data rather than free text.The authors developed a benchmark that assesses the ability of large language models to enact reasoning over Arabic tables. The authors tested various LLMs on table-based QA tasks and found that, while the LLMs had a strong understanding of text, they struggled with subordinating reasoning to approaches related to numerical reasoning and relationships between cells. The authors highlighted important obstacles in competing with enumerated hierarchical headings, numerical formats, and semantic table structures in the context of Arabic. Thus, by refining the initial template of unstructured passages to slots of structured data, it extends Arabic QA research space and presents an original evaluation domain to muster from the same LLM.

Paper [12] developed ArTrivia, a dataset generated by automatically collecting and filtering question–answer pairs from Arabic Wikipedia. The paper presented a pipeline for passage selection, automatic question generation, and human verification to ensure data quality. BERT and RoBERTa – both transformer-based models– were fine-tuned and evaluated to provide baselines. The authors noted that many Wikipedia-based corpora improved model generalization and domain coverage. ArTrivia presents a high-quality, open-domain Arabic QA dataset for established benchmarks and comparative evaluation across transformer architectures.

Paper [13] states that most of the existing QASs are based on a specific domain,which can limit the user's options. Also, Arabic QAS requests the user to provide in-depth information to be able to answer the user's query. This paper aims to address these limitations by developing an open-domain QAS that requires a minimum number of terms to answer the user query using deep learning techniques, especially quaternion long-short-term memory (QLSTM), multinomial Naïve Bayes (MNB), and embedding from language model (ELMo). The system includes three stages: data pre-processing, name entity relationship, and response retrieval. The proposed system was tested on two different datasets, the ARCD dataset and the TyDiQA dataset. The results of the study indicate that the system achieved high values in accuracy, recall, F1-score, MCC and Kappa. Despite the high metrics that the model achieved, the authors state that the performance of the model may be different in real-world applications, which leads future studies to enhance the performance of Arabic QAS.

Paper [14] introduces an innovative and cost-effective approach to train the AraT5 large language model (LLM) on a generated dataset. The paper addresses the limitation of finding annotated data to train QA models by using GPT-3.5 to generate Arabic question-answer pairs from AraSQuAD dataset. To effectively evaluate the performance of this model on the generated dataset, the same model was trained on a benchmarking dataset to compare the results. The results of this study indicate that the AraT5 trained on the generated dataset achieved a strong F1-score of 0.77, which is close to the F1-score of 0.88 from the AraT5 trained on the benchmarking dataset. These finding shows the promise of this approach and suggest that enterprises and institutions follow it when developing a QA system to minimize the cost. The author suggests future work to enhance this methodology on other models.

Paper [15] explains that question answering (QA) is the most challenging problem in the field of natural language processing (NLP), and the research in Arabic question answering is progressing slowly compared to the English one, due to the dataset availability. This study aims to evaluate and compare different pre-trained models, including AraBERTv0.2 large, AraBERTv2-base, and AraELECTRA, for the Arabic question answering task on four different datasets: the Arabic-SQuAD,ARCD, AQAD, and TyDiQA-GoldP. The evaluation experience starts with text pre-processing to clean the data, followed by a text segmentation process to meet the model's input specifications. The results of the study show that the AraBERTv0.2 large trained on Arabic-SQuAD dataset, and the AraELECTRA trained on the ARCD dataset, achieved the highest F1-score of 0.61 and 0.68,respectively. The study showed that pre-trained Arabic models can achieve strong results and improve the field of Arabic question answering tasks.

Paper [16] states that the Narrative question answering (QA) task includes responding accurately and in a relevant way to the user's questions, and this task is not well explored in the Arabic language, due to the dataset availability. This paper aims to address this limitation by building an Arabic Narrative question answering dataset. The dataset is created by translating both stories and question-answer pairs from the FairytaleQA dataset. The paper also introduces an Arabic Narrative question answering model. The model was built using the Ranker-Reader pipeline. This pipeline works by first ranking the most relevant paragraphs to the user's question using the AraBERT model, then reading the selected paragraphs using the mT5 and mBERT models to generate the correct answer.The paper presents both the dataset and the model to encourage research on Arabic Narrative tasks. Moreover, the study states a limitation on the reader that can only read three paragraphs due to the availability of GPU memory. Future studies can explore fusion-based techniques to overcome this limitation and process several paragraphs.

A. Gap Identification

After reading several papers, there are clear gaps in the field of question answering in Arabic NLP. Several papers, including [1],[3],[4], and [5], show that there is a lack of high-quality and large datasets. Other papers like [2] and [9] are only theoretical with no real experiments. Papers [7] and [8] use machine learning techniques that are limited in feature engineering and are computationally costly. Papers [10] and [12] are limited in comparing the proposed methods. Moreover, paper [13] shows a limitation in the method's performance in real-world scenarios, while [15] shows the limitations in mistranslation of the datasets. Papers [11] and [16] show a limitation in the model's performance in text reading.

In addition to these gaps, very few studies have provided a benchmarking framework that evaluates Arabic QA models across diverse datasets. This lack makes it difficult to understand how transformer models perform on different types of Arabic datasets. This gap highlights the need for an experimental evaluation study of different models, which this project aims to address.

IV. Data Acquisition and Preprocessing

A. Arabic-SQuAD

Arabic-SQuAD is a translation of a set of questions and passages from SQuAD 1.1 into Arabic [17]. This was done for both the question and answer pairs as well as the passages taken from Wikipedia in English. Since translations were done from the English language, it naturally has some translation noise, which resulted from using automated tools for translation. This dataset is designed for extractive question answering, where answers were extracted from the passage without requiring any form of generation, which aligns with the extractive form of the original SQuAD dataset.

B. ARCD (Arabic Reading Comprehension Dataset)

ARCD is a native QA corpus that has been constructed solely from Wikipedia articles written in Arabic, with a primary focus on reading comprehension. The size of the ARCD dataset is relatively smaller compared to Arabic-SQuAD, which could increase variance during training. Despite this limitation, ARCD contains valuable native Arabic content, which enhances its relevance for the model [18].

C. AQAD (Arabic Question Answering Dataset)

AQAD is a Wikipedia-based Arabic QA dataset that contains both answerable and unanswerable questions [18]. Since this issue can affect the performance of the models, we removed these pairs during preprocessing step.

D. TyDiQA-GoldP (Arabic subset)

TyDiQA-GoldP is a subset of the multilingual TyDiQA dataset, containing Arabic questions that were originally written by native speakers. Since TyDiQA-GoldP contains questions written by native Arabic speakers, it is free from translation noise, which is common in datasets that rely on machine translation.This makes TyDiQA-GoldP particularly valuable for extractive QA tasks, as it provides a clear definition of correct answers. This ensures that the model is trained on a clean, high-quality dataset [18].

TABLE I
ARABIC QUESTION ANSWERING DATASETS USED IN THIS STUDY

Dataset	Training Size	Testing Size
Arabic-SQuAD	38,885	9,459
ARCD	693	702
AQAD	4,108	1,151
TyDiQA-Gold (Arabic)	14,805	921

E. Data Preprocessing

Preprocessing is a critical step to ensure data compatibility with transformer-based architectures. In this work, we adopted a multi-stage preprocessing pipeline. First, all datasets were unified into a consistent SQuAD-style structure that consists of (id,question,context, and answer) to enable a shared processing workflow. After that, model-specific text preprocessing was applied following the steps described in the original MARBERTv2 and CAMeLBERT studies. Finally, the HuggingFace preprocessing and tokenization functions were used to convert the cleaned text into model-compatible inputs. Since text preprocessing alters character offsets, answer realignment method inspired by the original BERT preprocessing function was used to preserve correct answer positions.

In alignment with the MARBERTv2 paper [19], which was pre-trained on a massive dataset of 1billion Arabic tweets, our preprocessing focused on handling noisy social media text. Based on the paper methodology and our implementation, the following steps were applied. Removal of URLs, user mentions (@USER), and hashtags (#HASHTAG) to reduce non-semantic noise. Stripping of diacritics (Tashkeel) and normalization of whitespace to match the model's vocabulary.

Following the CAMELBERT paper guidelines [20], these specific steps were implemented. Removing the elongation character (Kashida/Tatweel) which is common in Arabic text but semantically redundant. Stripping all diacritics to unify the text representation, and removing extra whitespace to improve text presentation.

Due to the preprocessing steps, the alignment between some answers and their corresponding context became inconsistent. To address this problem, answer realignment was applied inspired by the answer realignment function used in original BERT-based question answering implementation. This method attempts to recover the correct answer spans by first performing exact matching on cleaned context, if the exact matching fails, fuzzy matching is performed to approximate the answer position [21].

F. Tokenization

We utilized the AutoTokenizer from the HuggingFace Transformers library to convert raw text into model-readable inputs. The tokenization process involved the following technical steps. The question and context were concatenated into a single sequence using special tokens: `[CLS] Question [SEP] Context [SEP]`. Followed by handling long contexts, since many passages in the datasets exceeded the maximum sequence length (e.g., 384 tokens), we applied a sliding window strategy (truncation with stride). This method splits long documents into overlapping segments, so we don't miss answers right where they split apart. Moreover, we used `return_offsets_mapping` to link each token to its place in the original text. So instead of characters, we could work out where tokens began and ended. That way, the dataset's 'Start Character' and 'End Character' became usable as 'Start Token' and 'End Token'. Needed this shift so the model learned from correct token spots [22].

V. MODEL DEVELOPMENT AND TRAINING

A. MARBERT Discription

The first model used in our experiments is the **MARBERTv2** transformer-based language model, which is specifically designed for Arabic. MARBERTv2 follows the BERT Base architecture and is pre-trained on an exceptionally large corpus of Arabic social media text, including approximately 1 billion Arabic tweets (around 128GB, equivalent to 15.6 billion tokens). This large and diverse dataset allows MARBERTv2 to effectively handle various Arabic dialects, informal writing styles, and noisy real-world text. MARBERT adopts the standard BERT-Base configuration, it has 12 transformer encoder layers and 768 hidden size and 12 attention heads, wordPiece vocabulary of approximately 100K tokens [19].

B. CAMeLBERT Discription

The second model used in our study is **CAMeLBERT**, a family of Arabic BERT-based models trained on different varieties of Arabic. CAMeLBERT is available in several variants CAMeLBERT-MSA for Modern Standard Arabic and CAMeLBERT-DA for Dialectal Arabic and CAMeLBERT-CA for Classical Arabic, which is used in this study, also CAMeLBERT-Mix for A combination of MSA, dialectal, and classical Arabic. CAMeLBERT follows the BERT-Base architecture it has 12 Transformer encoder layers and 768 hidden size also 12 attention heads and 30k WordPiece vocabulary [20].

C. Fine-Tuning Stage

The fine-tuning phase was performed following the HuggingFace question answering tutorial [22], ensuring that the choice of hyperparameters aligned with widely adopted best practices for Transformer QA. The training configuration was implemented using the HuggingFace Trainer API, which provided a complete pipeline for training, optimization, and evaluation.

First, both models' weights for question answering along with their tokenizers were uploaded from the HuggingFace model hub. After the uploading, all datasets were unified into a consistent structure to standardize the question, context, and answer fields. Each model then applied its own preprocessing pipeline, followed by the HuggingFace QA processing functions.

After the preprocessing steps, the training step started with the following configuration. The Trainer API uses AdamW by default with a learning rate of 2e-5, which is the value recommended by HuggingFace for stable fine-tuning. The models were trained for three epochs with a batch size of four, with a weight decay of 0.01 to prevent overfitting.

During the training process, both models compare their predicted start and end positions of the answer with the true answer position in the dataset. The difference between the actual positions and the predicted positions is measured for both start and end logits. After that, the models update their internal parameters to minimize this loss. After the training stage, a HuggingFace post processing procedure was applied to convert the models logits into the final test answer. Finally, the performance of the models was evaluated using Exact Match (EM) and F1- score.

VI. Evaluation and Analysis

The evaluation metrics used in this study are Exact Match (EM) and F1-score, which are commonly used in extractive question answering tasks.

A. Exact Match

Exact Match is a strict metric that returns one if the answer is predicted correctly and zero otherwise [23]. The formula shown in Eq. (1).

$$\mathrm{EM} = \frac{1}{N}\sum_{i=1}^{N} X_i \tag{1}$$

where $X_i = 1$ if the predicted answer equals the correct answer, and $X_i = 0$ otherwise.

B. F1-Score

F1-score provides a balanced measure of model performance by combining both precision and recall. It is particularly useful when partial matches between the predicted answer and the correct answer need to be evaluated. In extractive QA, the F1- score is calculated at the token level by comparing the overlap between the predicted answer span and the ground-truth span [23]. The F1-score is defined in Eq. (2).

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{2}$$

Since the HuggingFace configuration does not report validation accuracy during training, the following results are presented solely on the test set. Also, the original dataset splits were used in all experiments. The table below reveals the results of MARBERTv2 performance on the four datasets.

Dataset	Exact Match	F1-Score
	MARBERTv2	
TyDiQA-GoldP(Arabic)	77.52	86.53
ARCD	36.75	66.77
AQAD	22.06	35.88
Arabic-SQuAD	39.66	55.42

TABLE II
PERFORMANCE OF MARBERTv2 ON ARABIC QA DATASETS

The performance of MARBERTv2 varies considerably across the four datasets as shown in Table II. The model achieved its highest performance on TyDiQA-GoldP(Arabic), with an EM of 77.52 and an F1-score of 86.53, which indicates that the model identified the correct answer spans in this dataset. The performance of the model dropped on the ARCD dataset, where the model obtained an EM of 36.75 and an F1-score of 66.77, indicating that the model struggled to identify exact boundaries. The lowest performance obtained was on the AQAD dataset, with an EM of 22.06 and an F1-score of 35.88, indicating that the model faced challenges in this dataset. Lastly, on ARABIC-SQuAD, the model achieved moderate performance, with an EM of 39.66 and an F1-score of 55.42, which indicates partial success in extracting the correct answer.

Dataset	Exact Match	F1-Score
	CAMeLBERT	
TyDiQA-GoldP(Arabic)	73.50	83.81
ARCD	32.90	63.48
AQAD	18.85	30.08
Arabic-SQuAD	35.87	51.65

TABLE III
PERFORMANCE OF CAMeLBERT ON ARABIC QA DATASETS

Table III above shows the performance of CAMeLBERT on the same four datasets. On TyDiQA-GoldP(Arabic), CAMeLBERT showed its strongest performance, where it achieved an EM of 73.50 and an F1-score of 83.81, which shows that the model handled the dataset effectively and was able to locate the correct answer in many cases. The results of CAMeLBERT on ARCD decreased with an EM of 32.90 and an F1-score of 63.48, showing that the model struggled to identify the precise answer spans in ARCD. Moreover, the lowest results were on the AQAD dataset, where the model achieved an EM of 18.85 and an F1-score of 30.08 due to the higher difficulty of the dataset. On ARABIC-SQuAD, CAMeLBERT achieved an EM of 35.87 and an F1-score of 51.65. This moderate performance can be attributed to the dataset's translated nature, which made exact span extraction more difficult.

Dataset	C-EM	C-F1	M-EM	M-F1
TyDiQA-GoldP(Arabic)	73.50	83.81	77.52	86.53
ARCD	32.90	63.48	36.75	66.77
AQAD	18.85	30.08	22.06	35.88
Arabic-SQuAD	35.87	51.65	39.66	55.42

TABLE IV
COMPARISON OF CAMELBERT (C) AND MARBERTV2 (M) ON ARABIC QA DATASETS

When comparing the performance of both models, Both models follow a very similar performance pattern. First, both models achieved their highest results on TyDiQA-GoldP(Arabic) dataset, which can be because this dataset is a high-quality human annotated dataset. In contrast, both models achieved their lowest results on AQAD dataset although the unanswerable questions were handled effectively. This leads us to conclude that the AQAD is a challenging dataset for both models.

On ARCD and Arabic-SQuad datasets, MARBERTv2 acheived a better results in both F1-Score and Exact Match (EM). This result shows that although both Transformer models exhibit similar strengths and limitations when applied to extractive question answering in Arabic, MARBERTv2 outperforms CAMeLBERT on the four tested datasets, which can be explained by its pre-training on a very large and diverse corpus of Arabic social media text, allowing it to generalize more effectively across different question answering datasets.

VII. Conclusion (and Future Work)

To summarize, this paper contributes to advancing Arabic Question Answering by enabling a focused comparison of two contemporary Arabic transformer-based models MARBERTv2 and CAMeLBERT on representative Arabic QA datasets. The results of our work show that although both Transformer models exhibit similar behavior on the different datasets, MARBERTv2 achieved a better results in extracting the correct answers, likely due to its large-scale pre-training on a diverse corpus. Future work may further improve this project by using generative models (such as AraT5) to respond to user queries without relying on previously collected data to do so. On top of that, testing these models on purely dialectal datasets and exploring model distillation techniques could further enhance their applicability for real-time systems and non-standard Arabic content.

The source code for this project is publicly available at https://github.com/Sajedah25

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